

PARTS LIST

Semiconductors:

IC1	- 7805, 5V regulator
IC2	- P89V51RD2 microcontroller
IC3	- AT24C02 EEPROM
D1-D4	- 1N4007 rectifier diode
LCD	- 16x2 line LCD module
LED1	- 5mm LED

Resistors (all 1/4-watt):

R1	- 330-ohm
R2	- 10-kilo-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1000µF, 25V electrolytic
C2	- 0.1µF ceramic
C3	- 1 µF, 16V electrolytic
C4, C5	- 33pF ceramic

Miscellaneous:

X1	- 230V AC primary to 9V, 500 mA secondary transformer
S1-S17	- Push-to-on tactile switch
X _{TAL}	- 11.0592MHz crystal

and serial programming like in-system programming (ISP).

Data pins D0 through D7 of the LCD are connected to port pins P2.0 through P2.7 of IC2, respectively. Control pins register select (RS), read/write (R/W) and enable (E) are connected to port pins P3.5, P3.6 and P3.7, respectively. Preset VR1 is connected to pin 3 of the LCD for contrast control. Power-on reset is provided by the combination of resistor R2 and capacitor C3. Switch S17 is used for manual reset. An 11.0592MHz crystal along with two 33pF capacitors provides the basic clock frequency to microcontroller IC2.

A 16-key numeric keypad for address/data entry is connected to port P1 of the microcontroller. Here only eight pins are used for scanning and sensing the sixteen keys. The keypad is arranged in a 4x4 matrix. There are four scan lines, which are set in output mode. Four sensing keys are used as input lines to the microcontroller.

Memory. IC AT24C02 is an I²C-bus compatible 2kb EEPROM organised as 256x8-bit that can retain data for more than ten years. To obviate loss of the latest setting in case of power failure, the microcontroller can store the data in the EEPROM. The memory ensures that the microcontroller will read the last saved

data from the EEPROM when power resumes. Using SCL and SDA lines of the EEPROM, the microcontroller can read and write the data to/from AT24C02 memory.

Some of the conditions to communicate through the I²C bus are:

Clock and data transition. The SDA line is normally pulled high with an external device. Data on the SDA line may change only during SCL's low time period. Data changes during SCL line high time period will indicate a start or stop condition.

Start and Stop. A start condition is a high-to-low transition of SDA while SCL line is high. The stop condition is a low-to-high transition of SDA line while SCL line is high.

Acknowledge. All addresses and data words are serially transmitted to and from the EEPROM in eight-bit words. The EEPROM sends a zero to acknowledge that it has received each word. This happens during the ninth clock cycle.

To write data from the microcontroller to the EEPROM, the following interface protocol must be followed:

1. Send start condition
2. Send chip address byte (say A0H) with read/write '0' bit ('0' for write or '1' for read) followed by an acknowledge bit
3. Send internal memory address byte (where you want to write) followed by an acknowledge bit
4. Send data byte followed by an acknowledge bit
5. Optionally, send any further data bytes
6. Send stop condition

To read data from the EEPROM to the microcontroller, the following interface protocol must be followed:

1. Send start condition
2. Send chip address byte (say, A0H) along with read/write '0' bit ('0' for write or '1' for read) followed by an acknowledge bit
3. Send internal memory address byte (where you want to write) followed by an acknowledge bit
4. Send a start condition again (start repeated)

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Crisp System



 Arduino Kit	 Fpga Trainer Kit
 DSP Trainer	 Fingerprint Module
 Force Sensor	 Color Sensor
 GSM Modem	 Voice Recognition Kit
 MP3 Module	 Versa Pad
 RF Transceiver	 Wi-Fi Transceiver
 Accelerometer Sensor Module	 Thermoelectric Power Generator
 Fire Sensor	 Peltier Device
 Humidity Sensor Module	 Pulse Rate Sensor Module
 Voice Recording and Playback Module	 Power line Communication Module
 RFID Reader Module	 Gas Sensor Module

All the above Modules are Ready to Interface with any Micro controller

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Project Guidance also available